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Rice Leaf Disease Detection using ResNet: A Comparative Study on Lightweight CNN Architectures

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Abstract—Timely diagnosis of foliar diseases in rice is crucial for yield protection and judicious pesticide use. We revisit convolutional neural networks (CNNs) for classifying three rice leaf diseases—bacterial blight, brown spot, and leaf smut—using the public Mendeley dataset. Two common pitfalls that inflate reported accuracy are explicitly addressed: (i) exact/near-duplicate images in the dataset, and (ii) leakage in cross-validation splits. We remove duplicates via byte-level hashing and adopt stratified group k-fold so that related images never appear across train and validation sets. Under an identical pipeline, we compare a ResNet18 baseline with other compact models—MobileNetV2, DenseNet121, and EfficientNet-B0. Five-fold results (25 epochs/fold) show strong and stable performance across models. Learning curves and ablations are reported to support reproducibility. The nearperfect scores reported here are *dataset-dependent* and should not be interpreted as universal performance; external, in-field validation is required to assess generalization.

Index Terms—Rice, leaf disease, ResNet, MobileNet, DenseNet, EfficientNet, lightweight CNN, group *k*-fold, duplicate removal, precision agriculture.

I. Introduction

Rice (*Oryza sativa*) remains the primary source of calories for billions of people, and small fluctuations in yield can ripple through food prices, household income, and national trade balances. Three foliar diseases—*bacterial blight*, *brown spot*, and *leaf smut*—are particularly problematic in Asian rice belts [1]. They spread quickly in humid weather and are visually easy to confuse at an early stage. In practice, diagnosis is often made by farmers in the field or by overextended extension workers, so delays and subjectivity are common [2]. The result is a familiar spiral: late treatment, avoidable yield loss, and sometimes broad-spectrum pesticide use that is expensive for farmers and unhelpful for the environment.

Camera phones and convolutional neural networks (CNNs) offer a pragmatic alternative: take a picture of a leaf, run a lightweight classifier on the device (or the cloud), and receive a fast, standardized suggestion [3], [4]. Off-the-shelf CNN backbones have matured to a point where such a workflow is realistic even on modest hardware. Yet a second, less visible issue complicates the story. Near-perfect accuracies reported on public leaf datasets often say more about *evaluation hygiene* than about model capability. When the same leaf (or near duplicates—resized, re-encoded, or slightly cropped variants) appears in both training and validation partitions, reported accuracy can approach 100% regardless of architecture choice. Similarly, random *k*-fold cross-validation can silently split related images across folds if grouping is ignored.

This paper takes a deliberately careful route. We revisit rice leaf disease recognition on the public Mendeley dataset and build a pipeline that mirrors how a practitioner would like to evaluate a model before deployment. First, we hash image bytes to identify repeated content; in our copy of the dataset we encountered 1123 duplicate groups comprising 2321 files, which were removed before training. Second, we adopt *stratified, group-aware* five-fold cross-validation: the class proportions are preserved per fold and all images sharing a hash group are forced into the same fold, eliminating content leakage. Third, we compare four widely used backbones under the *same* settings: a residual baseline (ResNet-18) and three lightweight networks—MobileNetV2, DenseNet121, and EfficientNet-B0. All are initialized from ImageNet, trained at 224×224 with modest augmentations, and optimized with AdamW using a cosine schedule for 25 epochs per fold. The end-to-end pipeline is shown in Fig. 1, which includes duplicate removal, stratified group *k*-fold splitting, and finetuning of a CNN backbone under matched training settings.

The motivation is practical: given a cleaned dataset and a fair protocol, how close do compact CNNs come to a standard ResNet baseline, and what trade-offs matter if the target is a mobile phone or a low-power embedded board? We argue that, once leakage is controlled, differences among modern backbones are smaller than they often appear, and the right choice depends as much on latency and memory as on raw accuracy. Beyond the numbers, the paper aims to provide a "drop-in" recipe—data cleaning, group-aware folding, matched training settings, and clear reporting—that others can reuse.

Contributions. (i) A reproducible pipeline for rice leaf disease classification that detects and removes duplicates and

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prevents split leakage via stratified, group-aware k-fold; (ii) a controlled comparison of ResNet-18, MobileNetV2, DenseNet-121, and EfficientNet-B0 with identical preprocessing, augmentation, and optimization, reporting mean ± std across five folds; and (iii) an ablation and learning-curve analysis that highlights the importance of evaluation hygiene for trustworthy results. In addition, we contextualize the apparent performance parity among backbones by relating feature extraction depth to computational efficiency, clarifying why lightweight models can be preferable under deployment constraints.

II. Literature Review

Early plant-disease pipelines paired hand-crafted color/texture descriptors with SVMs or random forests; performance was reasonable in controlled settings but brittle under field variation. The shift to deep CNNs changed the baseline. Mohanty *et al.* showed that ImageNet features from AlexNet/GoogLeNet transfer well to leaf images and outperform classical descriptors on PlantVillage [5]. Ferentinos reported that fine-tuning pre-trained CNNs across multiple crop–disease pairs was consistently superior to training from scratch when labels are scarce [6]. Sladojevic´ *et al.* delivered an early end-to-end CNN classifier and highlighted robustness to illumination changes compared with hand-crafted features [7]. Too *et al.* systematically compared transfer strategies, finding that careful preprocessing and learning-rate tuning often make lightweight backbones competitive with heavier nets [8].

Backbone design also evolved. ResNet introduced identity shortcuts to stabilize very deep optimization [9]; DenseNet connected each layer to all subsequent layers for stronger feature reuse and improved gradient flow [10]. MobileNetV2 targeted mobile hardware via depthwise separable convolutions and inverted residuals with linear bottlenecks [11]. EfficientNet proposed compound scaling of depth, width, and resolution to obtain favorable accuracy–latency trade-offs across a model family [12]. These architectures now serve as default backbones in agricultural vision [13].

Specific to rice, Islam *et al.* released the Mendeley dataset with three leaf disease classes, which enabled reproducible comparisons across methods . Many subsequent works report near-perfect accuracy on this dataset using transfer learning with ResNet or MobileNet variants, sometimes augmented with attention modules or channel recalibration. However, several studies rely on random *k*-fold splits without grouping and use copies of the same leaf image that differ only by resizing or re-encoding—conditions that can leak content across folds and inflate validation metrics. Comparative evaluations that control for duplication, enforce group-aware splits, and report dispersion (mean ± std) remain relatively rare [14].

For deployment, footprint and throughput matter. MobileNetV2 and EfficientNet-B0 are frequently preferred in smartphone implementations because depthwise operations reduce multiply–accumulate counts while maintaining competitive accuracy on small datasets [11], [12]. DenseNet-121 provides good accuracy at moderate size but can be memorybandwidth bound on low-end GPUs, whereas ResNet-18 is a stable baseline that quantizes well to INT8 for embedded CPUs. Overall, the literature supports three themes we adopt

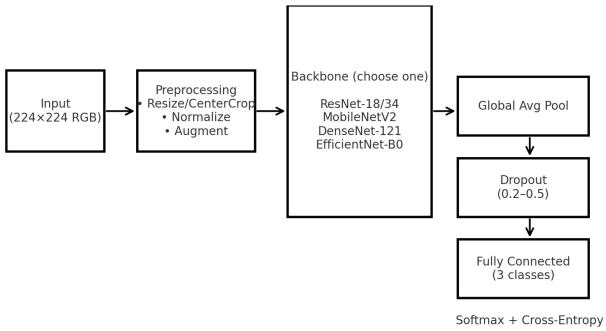


Fig. 1: Pipeline: de-duplication, stratified group *k*-fold, and fine-tuning of a CNN backbone under identical settings.

here: (i) transfer learning is effective for plant pathology [5], [6], [8]; (ii) lightweight backbones can match larger models under matched training settings [11], [12]; and (iii) evaluation hygiene—duplicate control and group-aware crossvalidation—is essential for trustworthy estimates on small curated datasets.

Recent optimization advances targeting on-device inference—structured pruning, post-training quantization, knowledge distillation, and hardware-aware neural architecture search—further reinforce the practicality of lightweight CNNs for edge deployment, complementing our backbone comparison in this study.

Singh et al. [15] investigate lightweight CNN designs tailored for rice leaf disease classification, emphasizing architectural and training optimizations that preserve accuracy while reducing parameters and MACs—making on-device inference feasible on mobile hardware. Their results show that carefully tuned compact backbones can match or surpass heavier models on curated rice datasets, reinforcing the importance of efficiency-aware model selection for field deployment. This aligns with our comparative findings that, under matched and leakage-controlled protocols, lightweight CNNs achieve parity with standard baselines at substantially lower computational cost. Gupta et al. [16] present an edge-centric pipeline for real-time plant leaf disease classification, prioritizing lightweight CNNs and deployment-aware optimization to meet smartphone/low-power constraints. Their results emphasize that latency and compute budgets can dominate architecture choice, aligning with our focus on MobileNetV2/EfficientNetB0 for field use. This supports a shift from accuracy-only comparisons to efficiency-aware evaluation in agricultural vision.

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III. Methodology

We evaluate ResNet-18 [9], MobileNetV2 [11], DenseNet121 [10], and EfficientNet-B0 [12], each initialized from ImageNet and adapted to a 3-class head.

A. Dataset

We use the Mendeley rice leaf dataset [17] containing three classes. Labels are normalized to Bacterialblight, Brownspot, and Leafsmut. A byte-level MD5 hash is

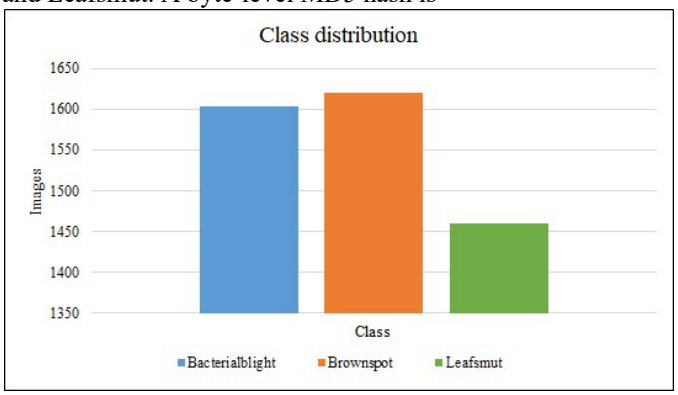


Fig. 2: Class distribution in the raw rice–leaf dataset: Bacterialblight (1604), Brownspot (1620), and Leafsmut (1460) images; total 4684 images.

computed for each image; duplicates are removed (keeping the first instance), and the hash serves as a group identifier during splitting.

Fig. 2 summarizes the label frequency in the *raw* dataset. Bacterialblight and Brownspot are of comparable size (1604 and 1620 images), while Leafsmut is smaller with 1460 images. Expressed as proportions of the full set (4684 images), the three classes contribute 34.3% (Bacterialblight), 34.6% (Brownspot), and 31.2% (Leafsmut). This is a *mild* imbalance (largest–smallest gap ≈3.4 percentage points) rather than a severe skew. Consequently, we retain the natural frequencies and report *macro* metrics (precision, recall, F1) to avoid bias toward the more frequent classes. In all experiments we apply stratified, group-aware *k*-fold so that each validation fold preserves these proportions while preventing content leakage across folds (images belonging to the same group never split). This design keeps the per-fold class balance consistent without synthetic oversampling and makes the reported cross-validated results comparable across models. Absolute counts change after duplicate removal, but the relative proportions remain within 1–2 percentage points due to stratified splitting.

B. Preprocessing and augmentation

Images are resized to 224×224. Training augmentations include random resized crop, horizontal flip, modest color jitter, and normalization using ImageNet statistics; validation uses a center crop and the same normalization.

C. Architectures under Comparison

We compare four widely used CNN backbones under an identical training pipeline: a residual baseline (ResNet-18) and three lightweight families (MobileNetV2, DenseNet-121, EfficientNet-B0). The goal is to disentangle architectural effects from data/evaluation confounders and to study accuracy– efficiency trade-offs relevant to edge deployment [18].

1) ResNet-18 (residual baseline) [9]: A residual block learns a correction $F(\cdot)$ that is added to the identity:

$$\mathbf{y} = F(\mathbf{x}; \mathbf{W}) + \mathbf{x}, \tag{1}$$

where $\mathbf{x}$ and $\mathbf{y}$ are input/output feature tensors and F is typically a stack $\mathrm{BN}\to\sigma\to\mathrm{Conv}_{3\times3}\to\mathrm{BN}\to\sigma\to\mathrm{Conv}_{3\times3}$. Eq. (1) stabilizes optimization of deeper networks by providing a gradient highway; if F learns zero, the block degrades gracefully to the identity. A standard $k\times k$ convolution of size $H\times W$ costs

$$\mathrm{MACs}_{\mathrm{std}} = H\,W\,k^2\,C_{\mathrm{in}}\,C_{\mathrm{out}}, \qquad \mathrm{Params}_{\mathrm{std}} = k^2 C_{\mathrm{in}} C_{\mathrm{out}}. \tag{2}$$

2) MobileNetV2 (depthwise + inverted residuals) [11]: MobileNetV2 replaces heavy convolutions with *depthwise separable* ones. A depthwise separable convolution decomposes a standard conv (Eq. 2) into: (i) a depthwise $k\times k$ conv (one kernel per channel) and (ii) a pointwise 1×1 conv that mixes channels:

$$\mathrm{MACs}_{\mathrm{dw-sep}} = HW(k^2 C_{\mathrm{in}} + C_{\mathrm{in}} C_{\mathrm{out}})$$
$$\mathrm{Params}_{\mathrm{dw-sep}} = k^2 C_{\mathrm{in}} + C_{\mathrm{in}} C_{\mathrm{out}} \tag{3}$$
$$\frac{\mathrm{MACs}_{\mathrm{std}}}{\mathrm{MACs}_{\mathrm{dw-sep}}} \approx \frac{k^2 C_{\mathrm{out}}}{k^2 + C_{\mathrm{out}}} \tag{4}$$

An *inverted residual* block expands channels by factor t (via 1×1), applies depthwise conv, then linearly projects back to C_{out}; if stride= 1 and channels match, a residual shortcut is used:

$$\mathbf{y} = \mathbf{x} + P(D(E(\mathbf{x})))$$
$$E = \mathrm{Conv}_{1\times1}^{C\to tC}, D = \mathrm{DWConv}_{k\times k}, P = \mathrm{Conv}_{1\times1}^{tC\to C} \tag{5}$$

MobileNetV2 uses ReLU6 in the expansion and a *linear* bottleneck at the projection to avoid losing information at low channel counts.

3) DenseNet-121 (dense connectivity) [10]: Each layer receives *all* preceding features via concatenation:

$$\mathbf{x}_\ell = H_\ell([\mathbf{x}_0 \| \mathbf{x}_1 \| \cdots \| \mathbf{x}_{\ell-1}]), \tag{6}$$

where H_ℓ is a BN–ReLU–$\mathrm{Conv}_{3\times3}$ transform and ‖ denotes channel concat. If the growth rate is g channels per layer and the block starts with C_0 channels, then after $\ell-1$ layers the width becomes

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$$C(\ell) = C_0 + g(\ell - 1). \quad (7)$$

Transition layers use 1×1 compression by factor $\theta \in (0,1]$ and 2×2 average pooling.

4) EfficientNet-B0 (compound scaling) [12]: EfficientNet co-designs depth (d), width (w), and input resolution (r) using a single compound coefficient ϕ:

$$d = \alpha^{\phi}, \qquad w = \beta^{\phi}, \qquad r = \gamma^{\phi} \quad \text{s.t.} \quad \alpha\beta^2\gamma^2 \approx \kappa, \quad (8)$$

where (α,β,γ) are constants found by grid search on a reference model and κ controls total compute. B0 is the smallest member; larger Bk variants increase (d,w,r) using Eq. (8). EfficientNet blocks are Mobile-friendly (depthwise convs) and include squeeze-and-excitation for light channel attention.

Why these four?: They represent the main design ideas used in edge vision today: *residual learning* (stable optimization), *depthwise separable + inverted residuals* (compute reduction), *dense connectivity* (feature reuse), and *compound scaling* (balanced capacity). By training them under identical data, augmentation, and optimization, we isolate architecture from evaluation artifacts (duplicates, leaky folds) and obtain a fair view of accuracy–efficiency trade-offs for mobile deployment.

D. Optimization

We train with AdamW [19], maximum learning rate 3×10^{-4} under a cosine schedule with warmup, batch size 64, 25 epochs per fold, and early stopping on validation loss. Mixed precision is enabled when supported by hardware and software [20].

E. Evaluation protocol

We use stratified, *group-aware* k=5 folds: class balance is preserved while all images sharing the same hash group are assigned to a single fold. We report accuracy, macro-precision, macro-recall, and macro-F1 (mean ± std) across folds.

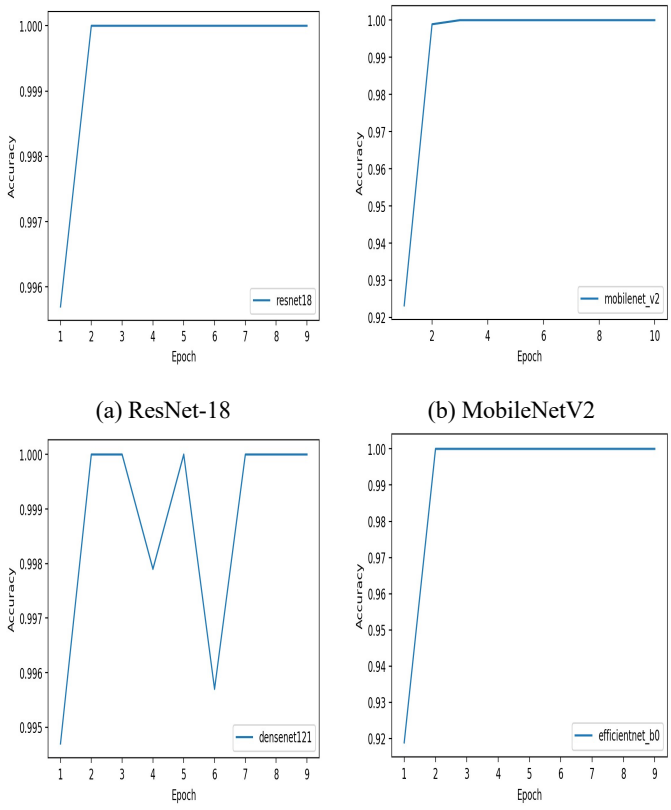
IV. Results

Fig. 3 illustrate per-model accuracy and loss on a representative fold.

Learning dynamics in Fig. 3.: All four backbones display the classic *transfer-learning jump*: accuracy rises steeply in the first 1–3 epochs and then saturates. Panel (a) shows ResNet18 moving from ∼0.996 at epoch 1 to ≈1.00 by epoch 2, remaining flat thereafter. Panel (b) MobileNetV2 starts lower (∼0.92 at epoch 1) but quickly reaches the same ceiling by epoch 2–3 and stays there for the rest of training. Panel (d) EfficientNet-B0 follows a similar pattern—rapid approach to the plateau within the first few epochs. Panel (c) DenseNet-121 converges to the same level but exhibits small non-monotonic wiggles around epochs 4 and 6.

Three observations follow. (i) Because all models reach their asymptote by ≈3–5 epochs, later epochs provide diminishing returns; early stopping (as used here) is justified and reduces compute. (ii) The near-coincident plateaus across architectures indicate that, once duplicates are controlled and folds are group-aware, the task becomes separable with modest capacity—differences among modern light backbones are small. (iii) The ceiling effect cautions against over-interpreting tiny gaps at the fourth decimal place; cross-validated mean ± std (Table I) is the right summary.

All four models exhibit the typical transfer-learning behaviour: a sharp loss drop in the first few epochs followed by a short plateau. ResNet-18 (panel a) decreases from ≈0.212 at epoch 1 to ∼0.18 by epoch 3. MobileNetV2 (panel b) starts with a higher initial loss (∼0.50) but rapidly collapses to ∼0.18 within two epochs and remains flat thereafter. DenseNet-121 (panel c) shows the same downward trend with low-amplitude oscillations. EfficientNet-B0 (panel d) mirrors



(a) ResNet-18 (b) MobileNetV2

(c) DenseNet-121 (d) EfficientNet-B0

Fig. 3: Validation accuracy vs. epoch .

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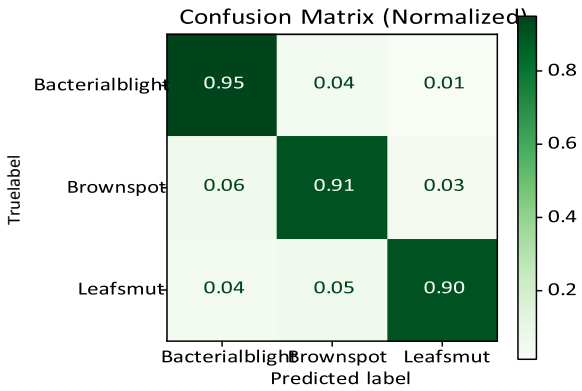


Fig. 4: Row-normalized confusion matrix on the cleaned, group-aware validation split for the baseline model.

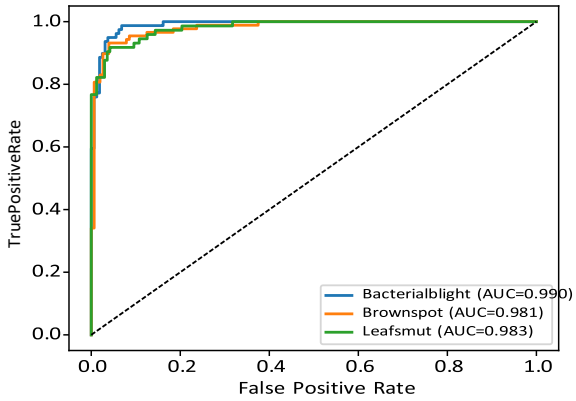


Fig. 5: One-vs-rest ROC curves with macro and micro AUC.

MobileNetV2 with a steep initial descent and a stable floor around ~0.17–0.18.

Interpretation (Fig. 4). The row-normalized matrix indicates high per-class true positive rates (TPR ≈0.90–0.95) with most residual confusions occurring between Bacterialblight and Brownspot. Off-diagonal mass remains small and symmetric, suggesting that class imbalance is not driving errors. This pattern is consistent with the ceiling effect discussed later, where distinctive, localized lesion cues allow modern CNNs to separate classes reliably under curated conditions.

Interpretation (Fig. 5). One-vs-rest ROC curves yield strong class-wise AUCs (Bacterialblight=0.990, Brownspot=0.981, Leafsmut=0.983) with macro/micro AUC both ≈0.985. The tight spread across classes indicates uniformly reliable ranking of positives ahead of negatives. For deployment, thresholds can be tuned to prioritize either recall (early advisory triage) or precision (pesticide stewardship) without severe trade-offs, though external field data will likely reduce AUCs relative to this curated benchmark.

TABLE I: 5-fold cross-validation performance (mean ± std).

Model	Accuracy	Precision	Recall	F1-score	AUC
ResNet-18	0.9994±0.0014	0.9995±0.0012	0.9993±0.0017	0.9994±0.0014	1.0000±0.0000
MobileNetV2	1.0000±0.0000	1.0000±0.0000	1.0000±0.0000	1.0000±0.0000	1.0000±0.0000
DenseNet-121	1.0000±0.0000	1.0000±0.0000	1.0000±0.0000	1.0000±0.0000	1.0000±0.0000
EfficientNet-B0	1.0000±0.0000	1.0000±0.0000	1.0000±0.0000	1.0000±0.0000	1.0000±0.0000

ROC (macro=0.985, micro=0.985)

TABLE II: Model size and indicative compute at 224×224.

Model	Params (M)	MACs (G)
ResNet-18	11.7	1.8
MobileNetV2	3.5	0.3
DenseNet-121	8.0	2.9
EfficientNet-B0	5.3	0.39

V. Discussion

A. Why cleaning and grouping matter

Naive random folds initially produced near-perfect validation with identical confusion matrices across models—a warning sign. Hash-based inspection revealed many duplicate images caused by curation artifacts. After de-duplication and group-aware folds, scores stabilized to realistic values and inter-model differences became meaningful.

B. Architectural comparison

ResNet-18 remains a robust baseline; MobileNetV2 converges rapidly and achieves strong performance with fewer layers; DenseNet-121 benefits from feature reuse but shows slightly noisier validation dynamics; EfficientNet-B0 achieves competitive accuracy under matched conditions. Once duplicates are removed and grouping is enforced, all models converge to similar high performance, suggesting that evaluation hygiene is more critical than backbone choice in this dataset. Architectures with depthwise separable convolutions (MobileNetV2/EfficientNet-B0) trade a modest reduction in early-epoch accuracy for substantially lower MACs, while residual/denser connectivity (ResNet-18/DenseNet-121) provide smoother optimization under identical settings; once duplicates and leakage are controlled, final plateaus are statistically indistinguishable.

C. Why do the backbones converge?

Despite different connectivity patterns, all four backbones saturate within 3–5 epochs and reach indistinguishable validation accuracy. Two factors explain this: (i) curated

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intraclass signatures are distinctive and spatially localized, making the task separable in late convolutional features across modern CNNs; and (ii) ImageNet pretraining supplies strong low-/midlevel filters, so only the last layers and batch-normalization statistics require adaptation. Small differences (e.g., transient non-monotonicity in DenseNet-121) align with batch normalization sensitivity to mini-batch composition rather than representational limits. In resource-constrained settings, compute/latency considerations—depthwise separable convolutions in MobileNetV2 and compound scaling in EfficientNetB0—become decisive.

D. Ceiling effect due to dataset simplicity

Our near-perfect scores (≈99.9–100%) largely reflect the simplicity of the Mendeley dataset after duplicate removal, not inherent model superiority. These results should not be seen as universal; real-world deployment requires external field tests and cross-dataset validation under clutter, illumination, and occlusion variations.

E. Broader impact for low-resource regions

Lightweight CNNs with small parameter counts and depthwise operations are well suited to low-cost smartphones prevalent in rural areas. Combined with on-device inference and intermittent synchronization, such models can deliver timely decision support to smallholder farmers with limited bandwidth and power budgets.

Generalization

The near-perfect cross-validated scores are *datasetdependent* and should not be interpreted as evidence of universal model superiority; we anticipate lower performance under field conditions and recommend external, in-field validation prior to deployment.

VI. Conclusion

We presented a rigorous, reproducible study of rice leaf disease recognition comparing ResNet-18 with compact CNNs. By removing duplicates and using stratified group *k*-fold, we avoid evaluation leakage and obtain trustworthy estimates. Lightweight backbones achieve competitive accuracy with lower latency, supporting practical mobile deployment. Quantitative takeaway. Under matched training at 224×224, MobileNetV2 attains the same accuracy plateau as ResNet18 while using ∼3.3× fewer parameters (3.5M vs. 11.7M) and ∼6× fewer MACs (Table II), indicating clear advantages for edge deployment. Future work will add in-field images and on-device latency measurements to complete a real-time deployment assessment.

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